Overfitting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Overfitting before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Overfitting trace and record the first step you can explain without help.

Use W2: Interpret a gap for **Overfitting** and model **capacity**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Overfitting explanation against the supplied facts before discussing it.

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Model A **training**96%, validation62%; model B **training**88%, validation79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Overfitting target to a trace and a justified conclusion.

A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Overfitting case and show the deciding evidence.

A scores100% train,**60%** validation; B scores85%,**80%** on the same sets.

Compute gaps and select by validation accuracy.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Model A has **training** accuracy 99% and validation accuracy **70%**; B has 88% and 84%. Compute each gap and discuss the **evidence**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 27 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Overfitting adds to Responsible Plant Classifier and keep the evidence.

Check Plant Classifier train-validation gaps and limit **complexity** before final test release.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Select from evidence

> Select from evidence. Apply the supplied Overfitting facts and keep your reasoning.

Choose among the three worked candidates and explain the **training** trap.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Interpret a gap

> Interpret a gap. Apply the supplied Overfitting facts and keep your reasoning.

Does a **training**-validation gap alone prove **overfitting**?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Propose a controlled change

> Propose a controlled change. Apply the supplied Overfitting facts and keep your reasoning.

Describe one fair next development comparison.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Overfitting facts and keep your reasoning.

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit a diagnosis

Limit a diagnosis. Apply the supplied Overfitting facts and keep your reasoning.

What extra **evidence** would you inspect before concluding a model memorized noise?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Overfitting

> See overfitting as a gap requiring investigation, not a label based only on model size. Explain the labelled step before checking the changed condition.

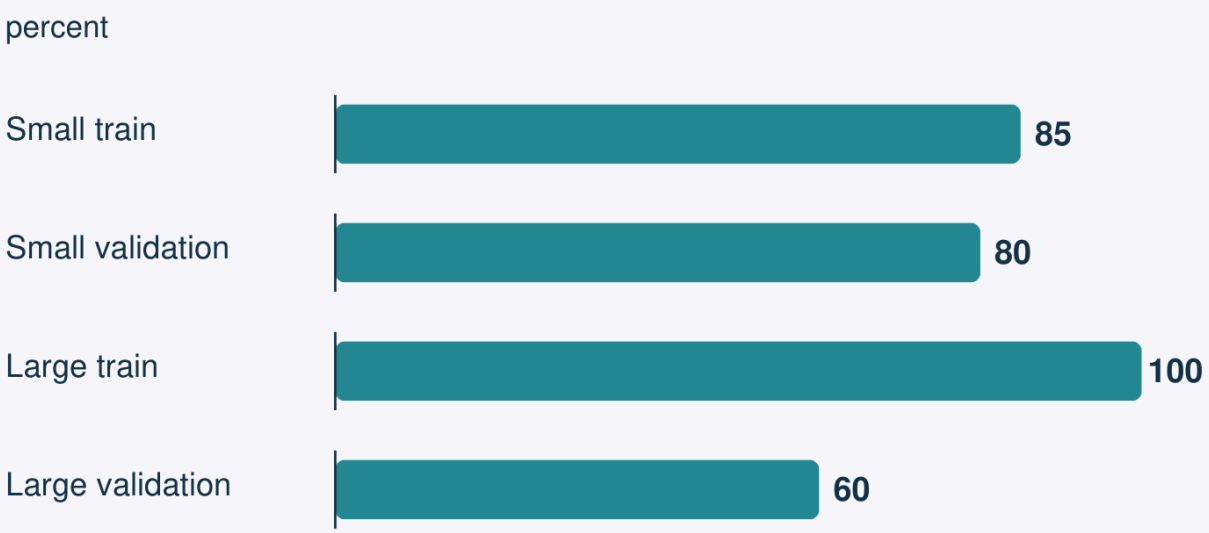
See **overfitting** as a gap requiring investigation, not a label based only on model size.

Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/27

Training and validation accuracy



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/27

Overfitting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gaps

Small gap=5 percentage points; Large gap=40 percentage points.



2. Select development candidate

Small has the higher validation accuracy despite its lower training fit.



3. Investigate large model

Check capacity, training choices and data mismatch without tuning on final cases.

CHANGE ONE THING

Large's validation accuracy rises to82% after a development revision.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See **overfitting** as a gap requiring investigation, not a label based only on model size.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check **capacity**, **training** choices and data mismatch without tuning on final cases.

1. Select development candidate

2. Compute gaps

3. Investigate large model

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Investigate large model Check **capacity**, **training** choices and data mismatch without tuning on final cases. Read the picture in order.

Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Large's validation accuracy rises to82% after a development revision. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Its gap is 18 percentage points; it now beats Small on this validation set.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its gap is 18 percentage points; it now beats Small on this validation set.

> Validation performance improved, but the **training**-validation gap is not a complete release rule.

Repair a missing building block

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

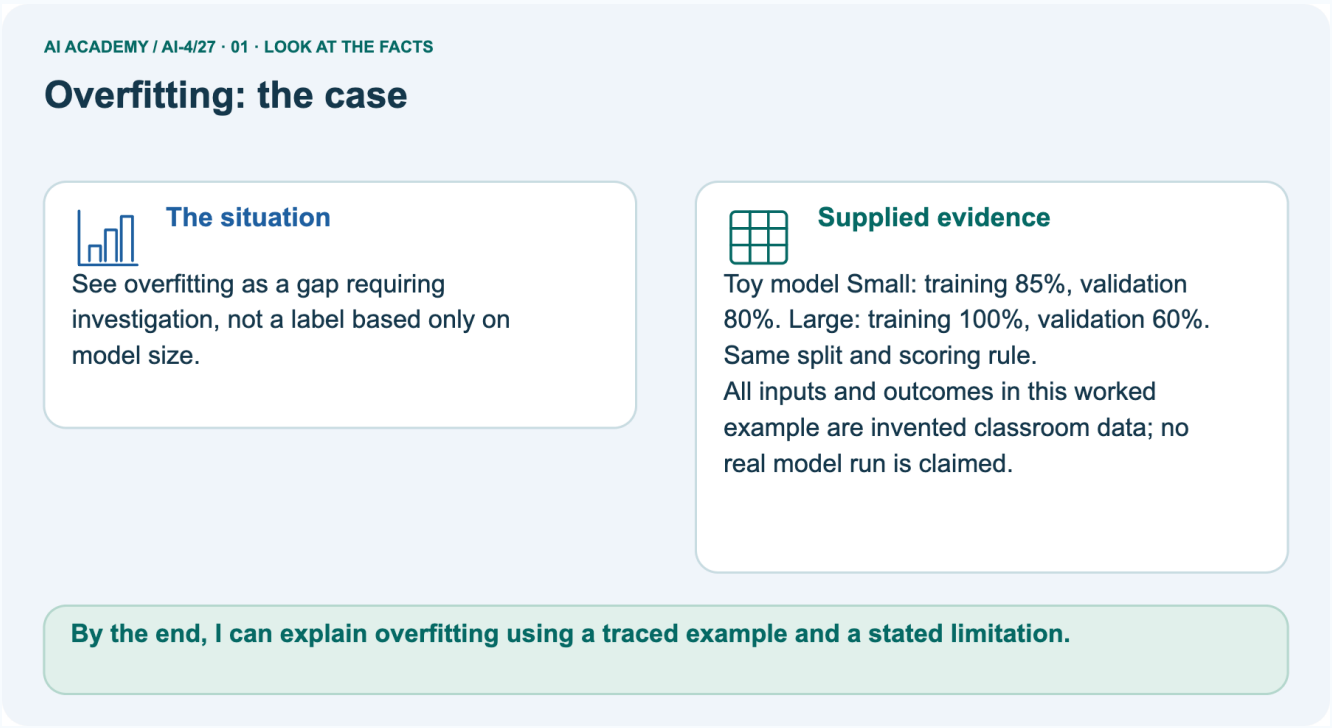
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Overfitting picture from its labels, then record a relevant limit.

01. Overfitting: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

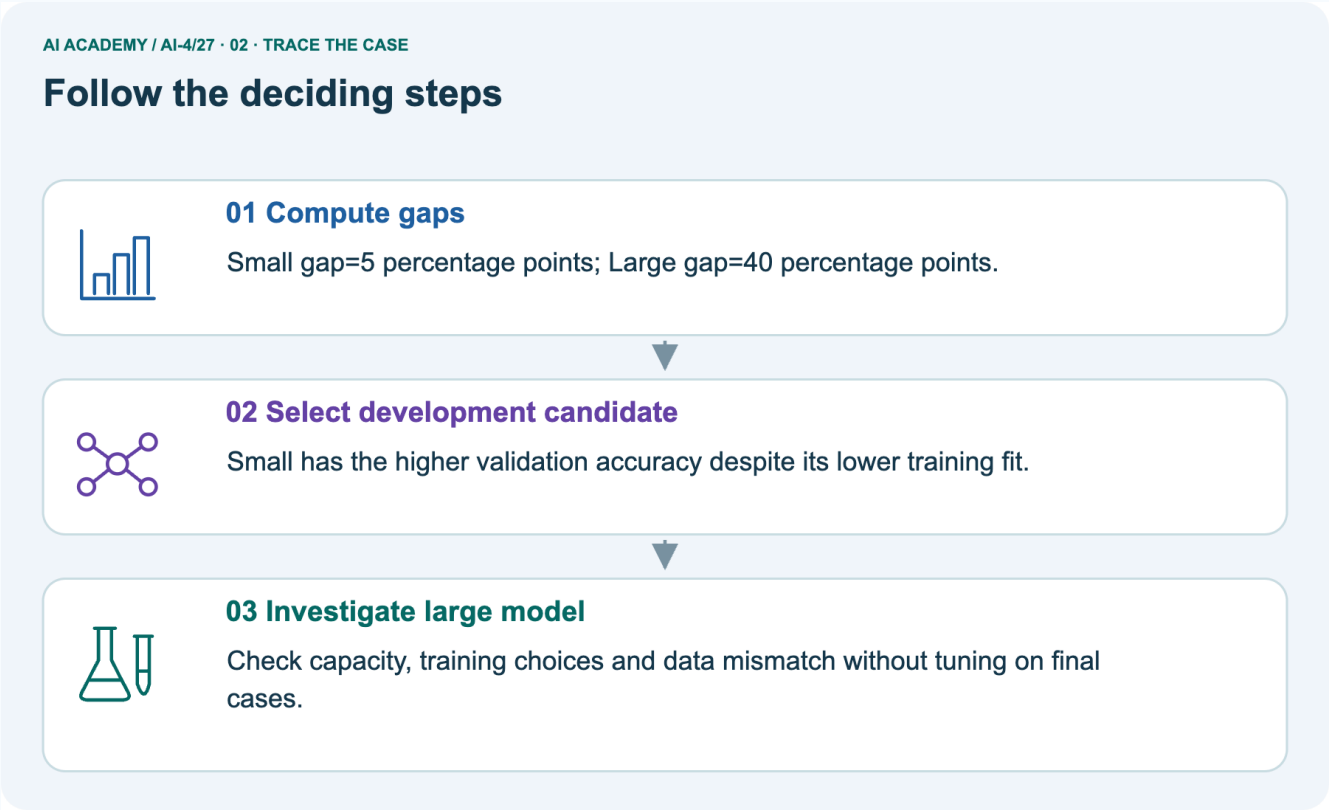
See **overfitting** as a gap requiring investigation, not a label based only on model size. Supplied facts: Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



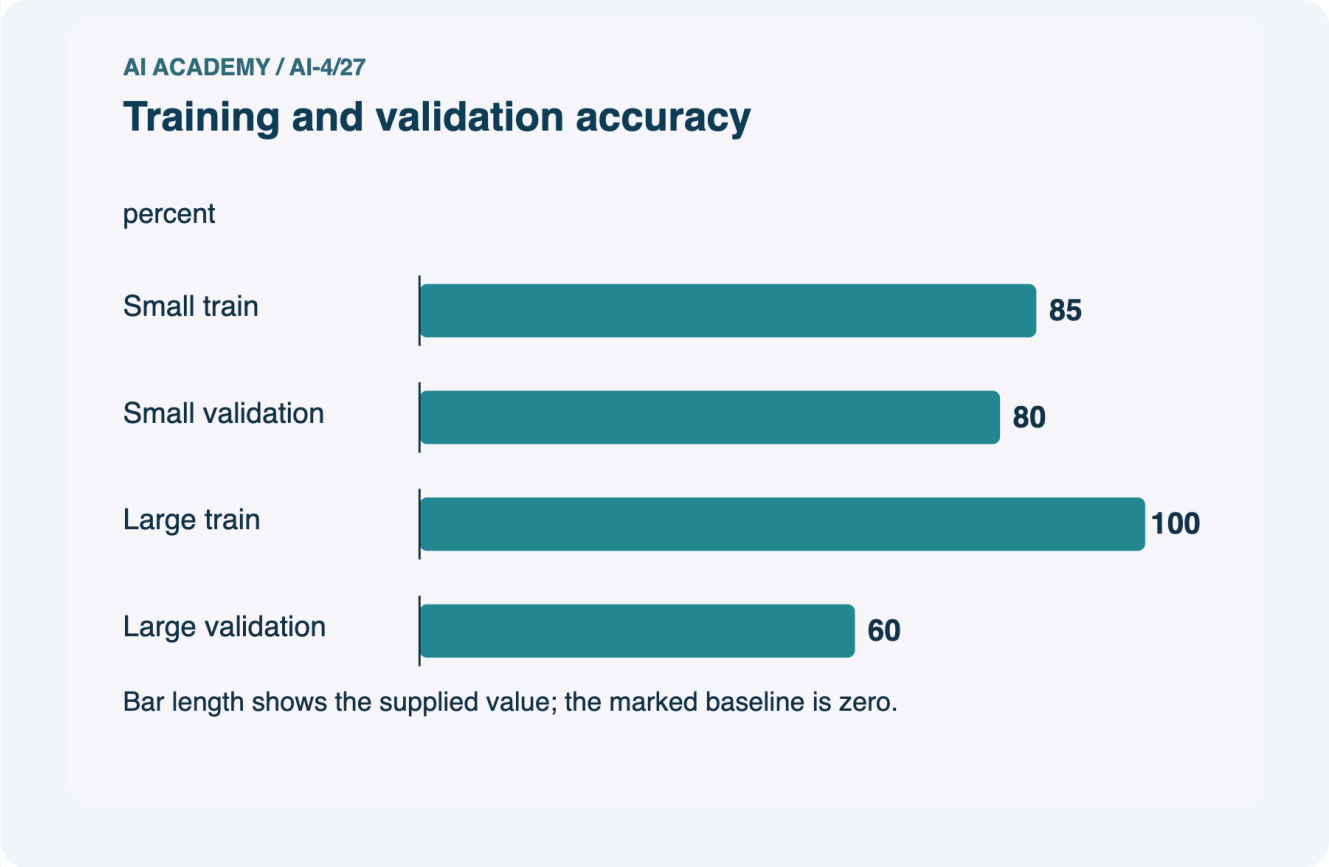
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gaps: Small gap=5 percentage points; Large gap=40 percentage points. → Select development candidate: Small has the higher validation accuracy despite its lower **training** fit. → Investigate large model: Check **capacity**, **training** choices and data mismatch without tuning on final cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Training and validation accuracy



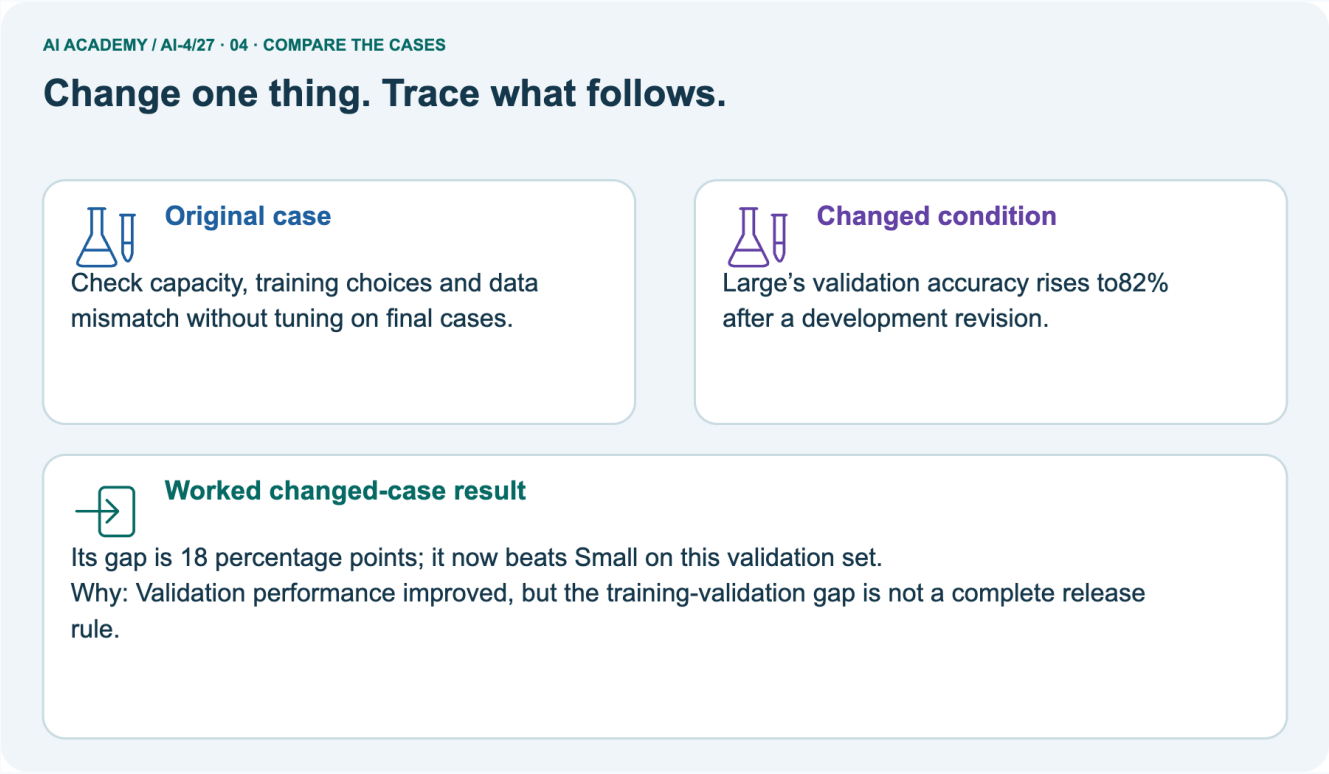
Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation accuracy. Units: percent. Small train: 85. Small validation: 80. Large train: 100. Large validation: 60.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

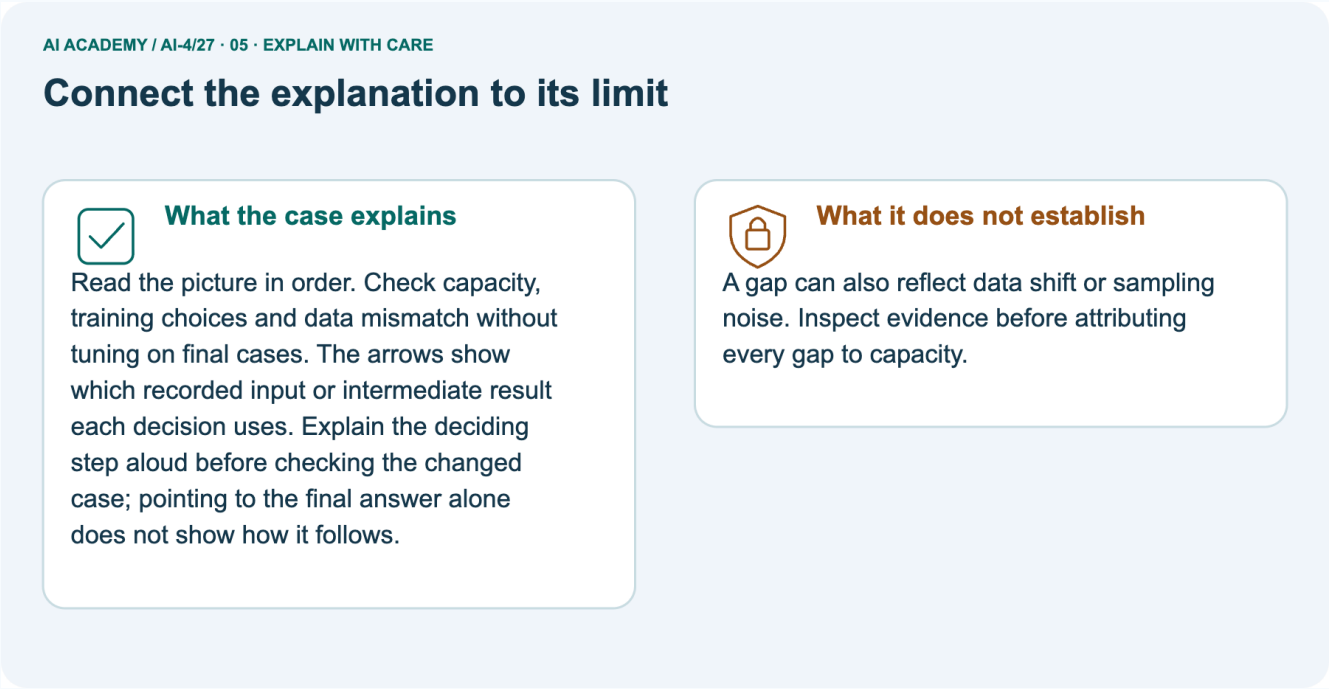
Original case: Check **capacity**, **training** choices and data mismatch without tuning on final cases. Changed condition: Large's validation accuracy rises to82% after a development revision.

Expected result: Its gap is 18 percentage points; it now beats Small on this validation set. Reason: Validation performance improved, but the **training**-validation gap is not a complete release rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A gap can also reflect data shift or sampling noise.

Inspect **evidence** before attributing every gap to **capacity**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.